

Use Case Specification: Book Event Ticket

1. Overview

- **Use Case ID:** UC-01
- **Use Case Name:** Book Event Ticket
- **Primary Actor:** Attendee

2. Preconditions and Postconditions

Preconditions:

- The attendee is logged into their account.
- The event is published, active, and has available tickets.

Postconditions:

- The ticket inventory is decremented by the purchased quantity.
- A digital ticket with a unique ID and QR code is issued to the attendee.
- A confirmation email with the ticket receipt is sent.

3. Main Success Flow

Step	Actor	Action Description
1	Attendee	Searches or filters active events.
2	System	Displays matching active events.
3	Attendee	Selects an event from the list.
4	System	Shows event details, pricing, and seat availability.
5	Attendee	Selects ticket quantity and clicks Checkout.
6	System	Verifies capacity and temporarily holds the requested tickets.
7	Attendee	Enters or confirms contact information.
8	Attendee	Reviews the order summary and total cost.

Step	Actor	Action Description
9	Attendee	Enters payment details and submits the order.
10	Gateway / System	Processes payment through the external gateway and approves the charge.
11	System	Permanently deducts the purchased tickets from remaining event capacity.
12	System	Generates a digital ticket with a unique ID and QR code.
13	System	Dispatches an email receipt with the digital ticket attached.
14	System	Displays the order confirmation screen.

4. Alternate Flows

4.1 Alternate Flow A: Insufficient Ticket Capacity (Sold Out)

Step	Actor	Action Description
6a	System	Detects that requested ticket quantity exceeds available remaining capacity.
6b	System	Displays an error message stating that the event is sold out or has insufficient tickets.
6c	Attendee	Adjusts the ticket quantity or returns to event listings.

4.2 Alternate Flow B: Invalid Contact Details

Step	Actor	Action Description
7a	System	Validates inputs and detects empty required fields or an invalid email format.

Step	Actor	Action Description
7b	System	Highlights the invalid fields and displays an error message prompting correction.
7c	Attendee	Corrects the contact information and proceeds to Step 8.

4.3 Alternate Flow C: Payment Authorization Fails

Step	Actor	Action Description
10a	Gateway / System	Payment gateway declines the transaction.
10b	System	Releases the held tickets back to available inventory.
10c	System	Displays a payment error message.
10d	Attendee	Prompted to retry with different payment details or cancel checkout.